



Combating Concept Drift with Explanatory Detection and Adaptation for Android Malware Classification

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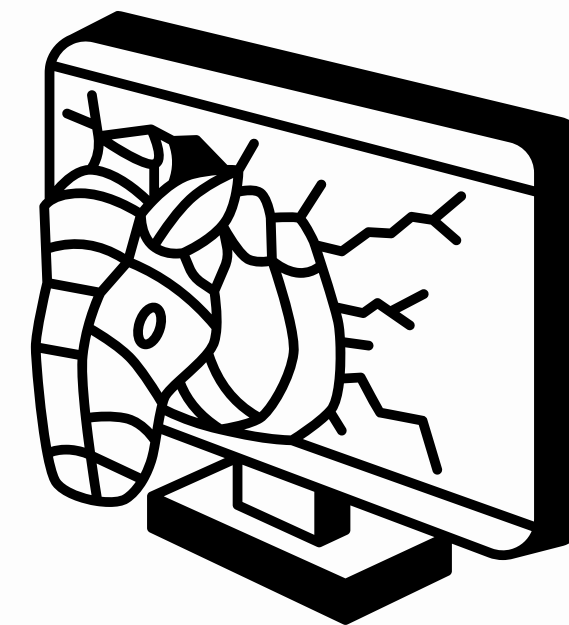
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研究問題

Android malware classifier 在部署後，惡意程式也會不斷演化。

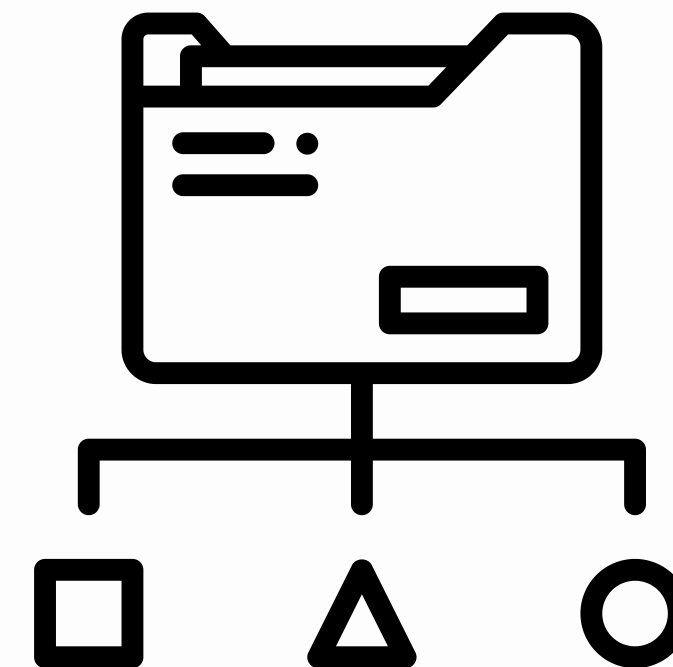
- 攻擊者會持續改變惡意程式的行為
- 新種惡意程式可能出現
- 舊模型只看過過去的資料，未見過的新樣本不一定能應付



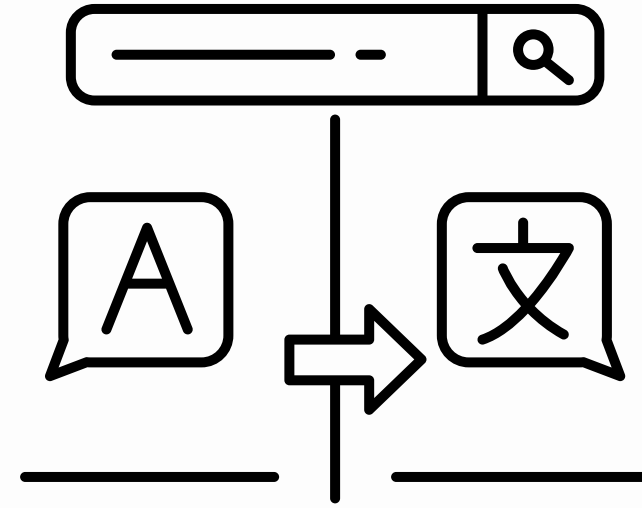
惡意程式分類器需要能夠適應新的威脅。

分類錯誤的影響

- 無法判斷 malware family
- 無法採取正確防禦策略
- 資安分析人員需要花更多時間人工判斷



分類器的工作



輸入

APK 中抽取出的特徵

- permissions
- API calls
- opcode sequences
- package call pairs
- suspicious behaviors
- etc

輸出

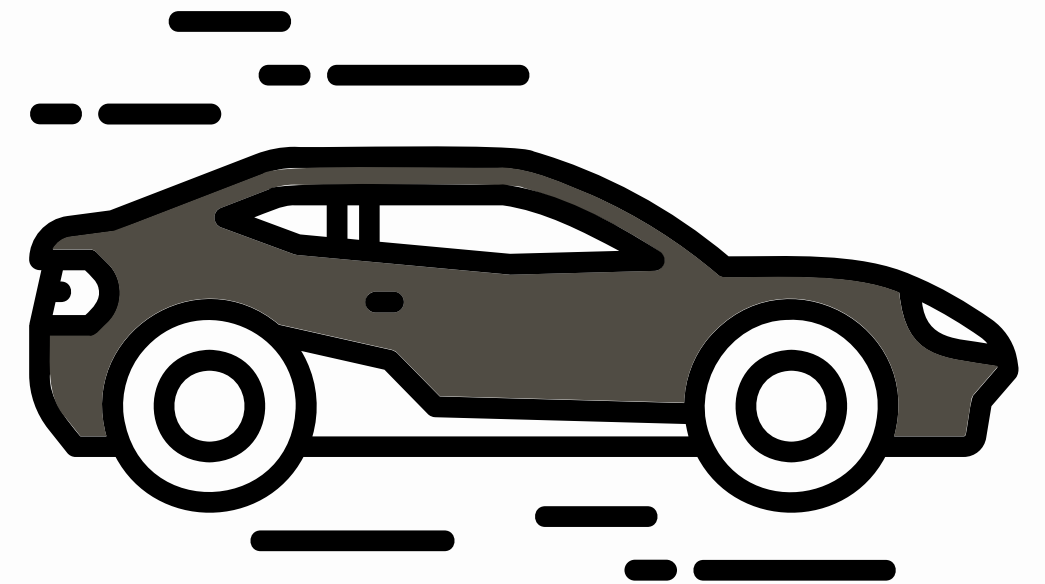
家族標籤

- FakeInstaller
- DroidKungFu
- GhostCtrl
- Xavier
- etc

CONCEPT DRIFT

資料的統計特性或資料與標籤之間的關係，會隨著時間改變，所以過去模型學到的分類規則可能不再適用。

e.g. 某些 API 行為模式通常代表 A 類，新的惡意程式改用不同 API 或不同隱藏方式，結果模型仍用舊規則判斷，因此分類錯誤。



第一類： INTRA-CLASS DRIFT

同一個家族內部分支出新的變種。

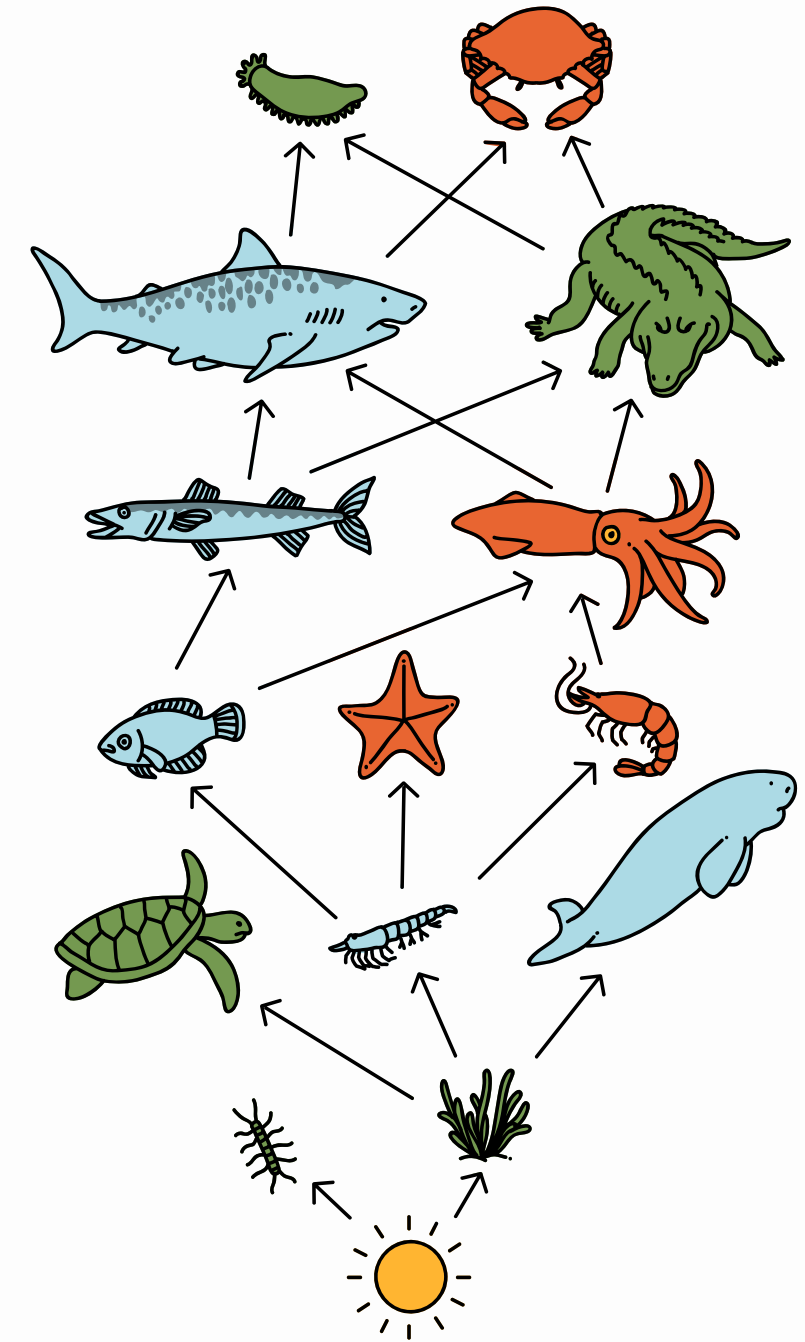
家族沒變但行為或特徵改變，導致模型可能仍然認不出新的變種。

例子: Xavier malware



第二類：INTER-CLASS DRIFT

指出現了過去訓練資料中沒有的新家族，也是本篇論文著重解決的問題。



現有處理方式：ACTIVE LEARNING

1. Detection

找出可能發生偏移的新樣本

2. Adaptation

把這些樣本交給專家標註，再餵給模型訓練。

偵測器學到的距離或異常分數，不一定符合分類器真正依賴的模式！
模型只使用 label，沒有使用專家的分析理由。需要更多人工標註資料，更新成本高。

DRIFT DETECTION 目前問題

1. Detector and classifier may learn different patterns

- Detector learns distance or anomaly scores
 - Classifier uses its own decision rules
- Inefficient drift detection

2. Dependence on training data

- nearest neighbor search
 - uncertainty estimation
- storage overhead, security concerns

DRIFT ADAPTATION 目前問題

- Current methods use human labels
 - But ignore human reasoning, behavioral explanations
- **Information loss, higher labeling cost**

HOW TO ADDRESS

Detection

Use classifier-guided concept reliability

- Better model-sensitive detection
- No training data dependency

Adaptation

Allow experts to revise labels, behavioral concepts

- Preserve expert knowledge
- Better model updating

D R E A M

DRift detection and Explanatory Adaptation for Malware classification

Core idea: Integrate classifier + expert knowledge

Two components:

- Model-sensitive Detection
- Explanatory Adaptation

DREAM 基本架構

Explanation + Behavioral concepts + Human knowledge

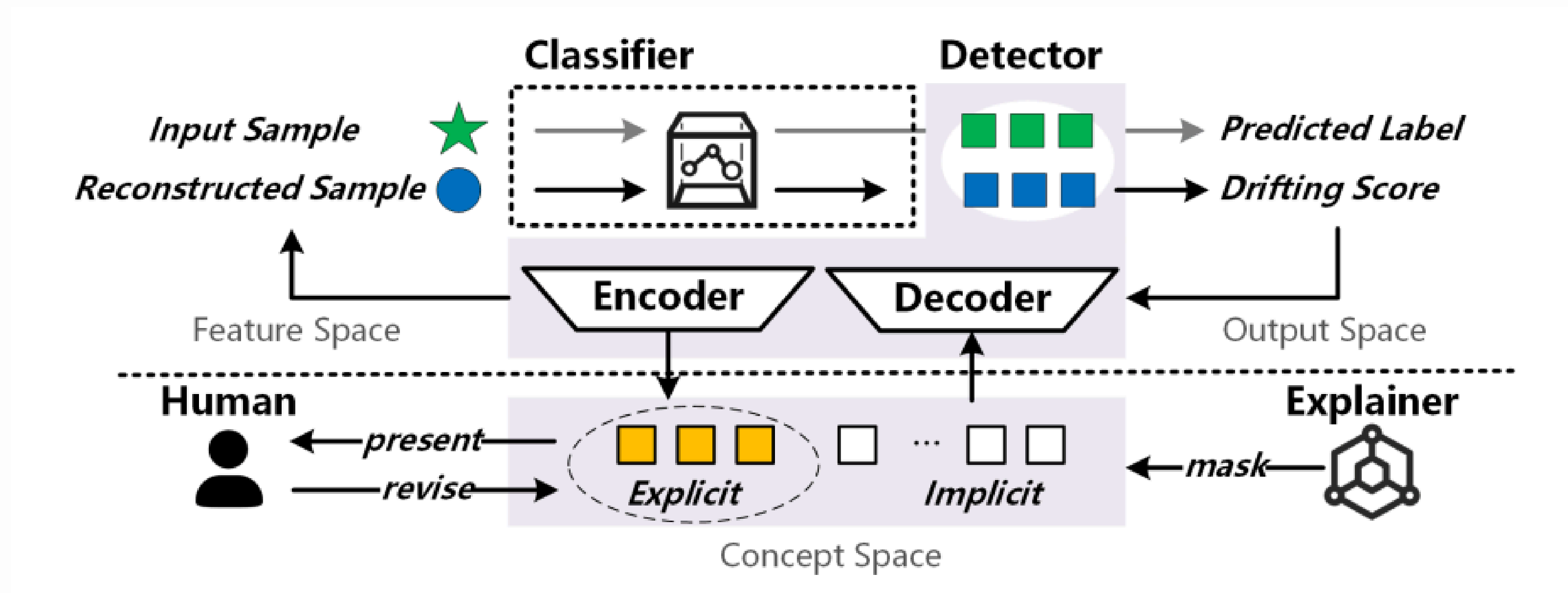
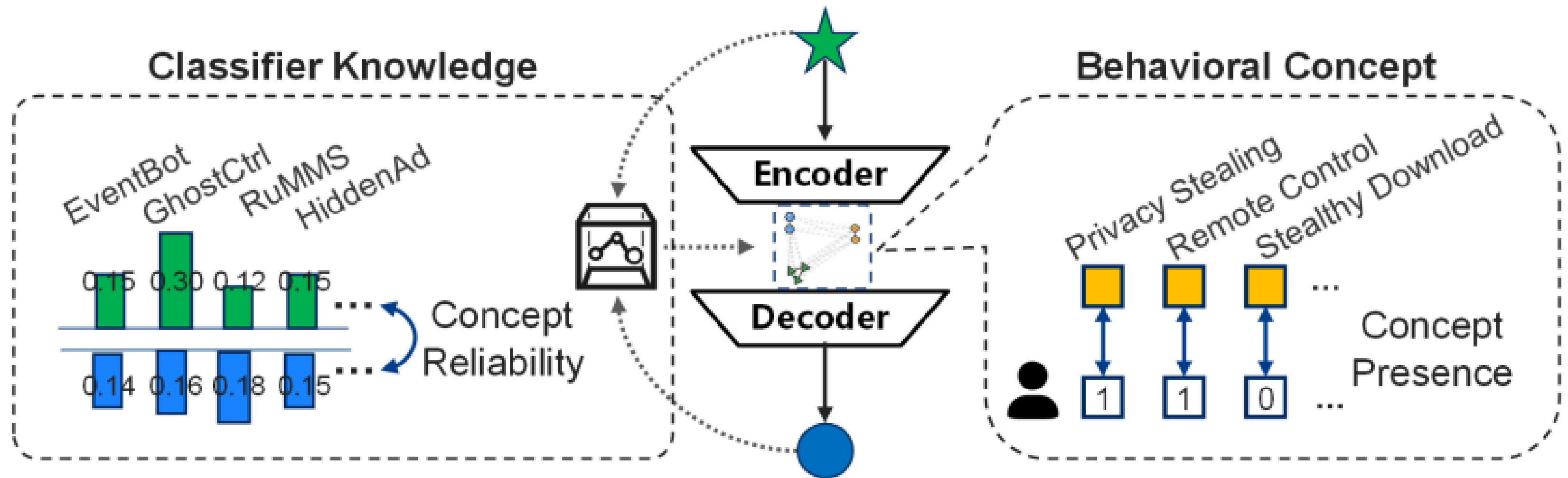


Figure 3: Overview

SOME TERMS

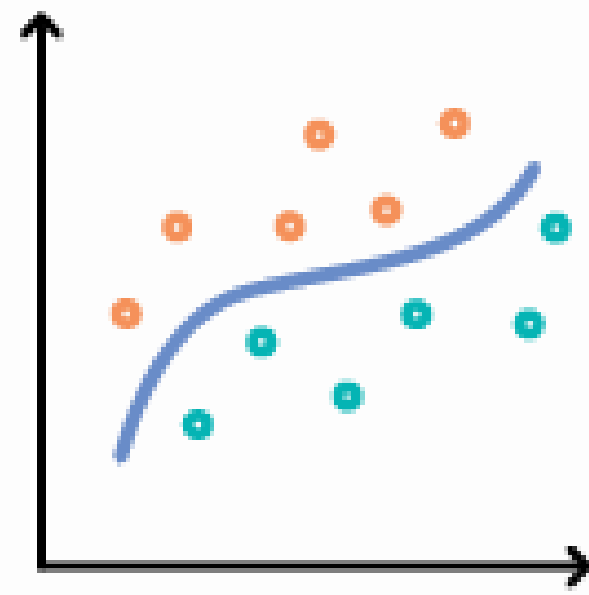
- **Drift explainer:** explains why a sample is identified as **drifting** and presents it in terms of interpretable malware behaviors rather than simple feature differences.
- Feature-based → Concept-based
- **Explicit concepts:** remote control, privacy stealing, ransom, stealthy download
- **Implicit concepts:** hidden behavior learned by model

Model Sensitive Concept Learning

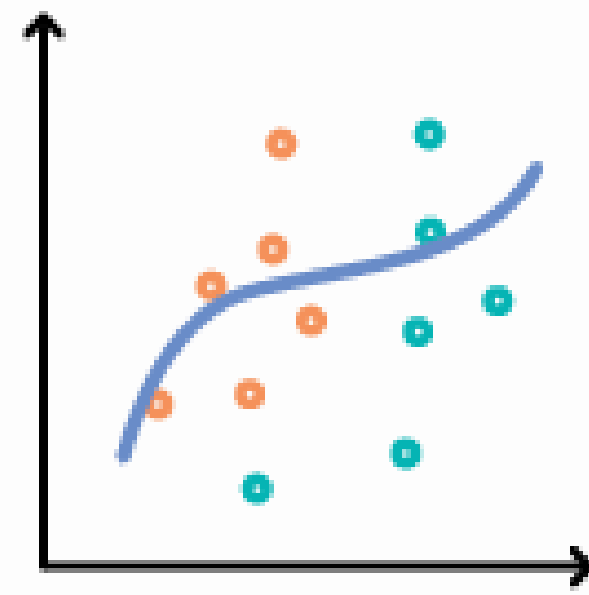


Design Insights

小結



Original Data
(at time t)



Feature Drift
(at time $t+1$)

Android malware evolution



Concept Drift



Existing Active Learning
(limitations)



DREAM



better detection + adaptation
with lower human cost

THANK YOU